
Natural Korean Sentence Generation from KSL Gloss Sequences using KoBART

Korea University COSE461 Final Project

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Abstract

Generating natural Korean sentences from Korean Sign Language (KSL) gloss sequences remains a challenging task because gloss representations often omit grammatical information and differ structurally from spoken Korean. While previous sign language studies mainly focus on video-based sign recognition and gloss extraction, relatively less attention has been given to the gloss-to-text generation stage. This project investigates the effectiveness of a pretrained Korean sequence-to-sequence model, KoBART, for converting KSL gloss inputs into fluent Korean sentences.

The proposed approach formulates gloss-to-text generation as a sequence-to-sequence translation problem. KoBART was fine-tuned using the GKSL3k dataset containing paired gloss and Korean sentence samples. To reduce sentence-type ambiguity, additional intent tags, [STATEMENT] and [QUESTION], were introduced into gloss inputs during preprocessing. The model was trained and evaluated using BLEU, ROUGE-L, and BERTScore metrics together with qualitative output analysis.

Experimental results demonstrate that the proposed model successfully generates understandable and semantically accurate Korean sentences from gloss inputs, achieving a BLEU score of 66.46, ROUGE-L F1 score of 0.7253, and BERTScore F1 score of 0.9231. Qualitative evaluation further shows that the model performs particularly well on short and frequently occurring expressions while preserving sentence structure and grammatical meaning. These findings indicate that pretrained Korean language generation models provide an effective and practical approach for Korean gloss-to-text translation.

1 Introduction

Sign language translation has become an important research area in natural language processing and accessibility technologies. In Korean Sign Language (KSL), gloss representations are commonly used as an intermediate textual form to describe sign language expressions. However, gloss sequences are structurally different from natural Korean sentences and often omit grammatical elements such as particles, tense markers, and sentence endings. As a result, gloss expressions are difficult for

general users to understand directly, making gloss-to-text generation an important task for improving communication accessibility.

Many previous sign language translation studies focus on video-based sign recognition, where hand movement, body posture, and joint information are used to recognize sign words or gloss labels. While these approaches have shown strong performance in recognizing isolated sign words, generating natural Korean sentences from gloss sequences remains challenging because sentence reconstruction requires semantic understanding and grammatical generation.

Recent advances in pretrained transformer models have significantly improved natural language generation tasks such as machine translation and text summarization. Among these models, KoBART is a Korean pretrained encoder-decoder transformer model optimized for Korean text generation. Its strong Korean language understanding and generation capability make it a promising approach for gloss-to-text generation tasks.

In this project, we focus specifically on converting KSL gloss sequences into natural Korean sentences using KoBART. To reduce sentence-type ambiguity, intent tags such as [STATEMENT] and [QUESTION] are added to the gloss input to explicitly indicate sentence form. The model is trained using a parallel gloss-sentence dataset and evaluated using BLEU, ROUGE, and BERTScore metrics. Through this project, we investigate the effectiveness of pretrained Korean language generation models for Korean gloss-to-text translation.

2 Related Work

Recent studies on sign language translation have mainly focused on recognizing sign language from video inputs using deep learning techniques. Many approaches use visual information such as hand movement, body posture, and joint coordinates extracted from sign language videos to predict sign words or gloss labels. These studies commonly employ convolutional neural networks (CNNs), recurrent neural networks (RNNs), graph-based networks, and transformer architectures for sign recognition tasks.

Shin et al. [1] proposed a dynamic Korean Sign Language (KSL) recognition system using pose estimation and attention-based neural networks. Their approach extracts hand and body joint information from sign language videos and constructs a graph representation for KSL recognition. The model applies graph convolutional network (GCN)-based feature extraction, channel attention, and CNN-based classification modules to predict sign words or gloss labels.

Another study by Shin et al. [2] proposed a transformer-based multi-branch network combining CNN and transformer architectures for KSL recognition. Their method extracts both local visual features and long-range dependency information from sign language inputs, then combines the learned features for classification. The model achieved strong performance on Korean Sign Language datasets, demonstrating the effectiveness of combining CNN and transformer-based representations for sign recognition.

Recent pretrained language models have also shown strong performance in Korean natural language generation tasks. KoBART [3] is a Korean pretrained encoder-decoder transformer model developed by SKT-AI for Korean text generation tasks. KoBART has been applied to various Korean NLP tasks including chatbot generation, text summarization, Korean-English translation, LegalQA, and question generation, demonstrating strong Korean language understanding and generation capability.

Kim and Cho [4] compared KoBART and ByT5 for Korean-to-KSL gloss translation using a Korean-KSL parallel corpus. Their study showed that KoBART significantly outperformed the byte-level ByT5 model in gloss generation quality, demonstrating the effectiveness of subword-based transformer models for KSL translation tasks. While their work focuses on Korean sentence-to-gloss generation, our project addresses the reverse task of generating natural Korean sentences from gloss sequences.

Unlike previous studies that mainly focus on sign recognition from video, this project focuses on the gloss-to-text generation stage after gloss extraction. Our work uses KoBART to convert gloss sequences into natural Korean sentences instead of predicting isolated sign words. In addition, intent tags such as [STATEMENT] and [QUESTION] are introduced to reduce sentence-type ambiguity during sentence generation.

3 Approach

This section outlines the neural network architecture and optimization mechanics developed to convert Korean Sign Language (KSL) gloss matrices into grammatically correct text. By operating directly at the tokenized linguistic interface, our system isolates the translation problem to a text-to-text sequence generation pipeline, bypassing computer-vision bottlenecks entirely.

The end-to-end processing pipeline maps inputs through four sequential processing layers, as illustrated in Figure 1.

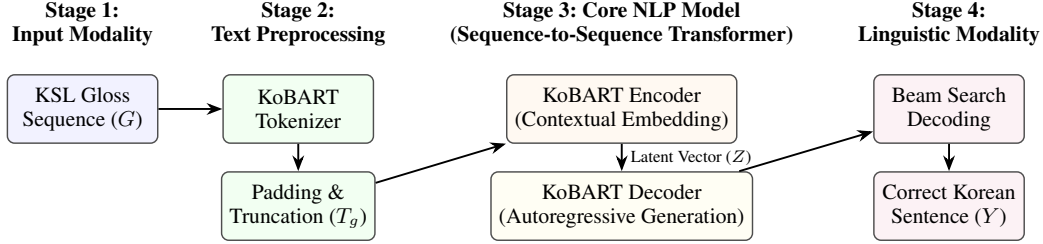


Figure 1: Architecture framework of the proposed system focusing entirely on NLP-driven text-to-text generation starting from structured KSL Gloss inputs.

3.1 KoBART-based Sequence-to-Sequence Model

The translation module leverages KoBART [3], a variant of the Bidirectional and Auto-Regressive Transformer (BART) architecture [5] pre-trained specifically on large-scale Korean corpora. The architecture incorporates a bidirectional encoder that reads ungrammatical, unordered sign gloss entries to formulate a comprehensive contextual representation, combined with an autoregressive decoder that predicts natural token steps from left to right.

Formally, let the input sequence of KSL gloss symbols be represented as a vector:

$$G = (g_1, g_2, \dots, g_n)$$

The raw tokens are mapped to discrete subword IDs via an optimized byte-level Byte-Pair Encoding (BPE) configuration, yielding the initialized tensor $T_g = \text{Tokenizer}(G)$. The tokenized array propagates through the multi-layer transformer blocks of the encoder to build a continuous latent representation matrix:

$$Z = \text{Encoder}(T_g)$$

Using the latent context vector Z , the decoder generates the distribution over the target natural language tokens:

$$Y = (y_1, y_2, \dots, y_m)$$

The objective of the model is to maximize the cumulative conditional probability distribution across the entire target space, which factors autoregressively as:

$$P(Y | G) = \prod_{t=1}^m P(y_t | y_{<t}, Z)$$

where y_t corresponds to the localized categorical probability vector across the shared model vocabulary size $|V|$, and $y_{<t}$ encapsulates all historically emitted target tokens.

3.2 Loss Optimization and Decoding Policies

To eliminate sentence-type ambiguity during sequence generation, we employ a prefix target conditioning strategy inspired by [6], where functional intent indicators are prepended to the input token sequence.

During the training phase, the parameter array weights θ are updated end-to-end. Optimization is driven by minimizing the standard categorical cross-entropy loss function over the tokens of the sequence-to-sequence generation dataset D :

$$\mathcal{L}(\theta) = - \sum_{i=1}^{|D|} \sum_{t=1}^{m_i} \log P(y_t^{(i)} | y_{<t}^{(i)}, G^{(i)}; \theta)$$

During inference, direct greedy decoding strategies are highly susceptible to local minima, structural loops, and exposure bias. To improve output fluency and structural accuracy, sequence derivation is calculated via a specialized Beam Search decoding strategy [7]. Instead of sampling a single maximal candidate at step t , the system tracks B active running hypotheses with the highest length-penalized log-likelihood scores, calculated as:

$$\text{Score}(y_1, \dots, y_t) = \frac{\sum_{\tau=1}^t \log P(y_\tau | y_{<\tau}, Z)}{t^\alpha}$$

where α represents the empirical length-penalty scaling exponent. Following Freitag and Al-Onaizan [7], this decoding approach balances search exploration and generation runtime while reducing repetition compared with greedy decoding.

3.3 Copying-Based Control Baseline

Previous sequence-to-sequence Sign Language Translation (SLT) studies emphasize the importance of transforming structured sign representations into natural language outputs rather than directly preserving input structure [8]. Motivated by this observation, a pass-through copying strategy was considered as a conceptual reference to illustrate the difference between direct gloss transfer and generative translation.

In this simplified setting, the output is treated as an identity mapping:

$$Y = G$$

Unlike the proposed KoBART framework, this conceptual baseline does not perform grammatical reconstruction, particle generation, or sentence restructuring. Table 1 summarizes the conceptual differences between both approaches.

Table 1: Architectural and structural comparison between the Baseline approach and the Proposed KoBART framework.

Metric / Component	Copying Baseline	Proposed KoBART Framework
Input Structure	Raw KSL Gloss (G)	Raw KSL Gloss (G)
Tokenizer Requirements	Identity / None	Pretrained BPE Tokenizer
Internal Layers	Pass-Through Matrix	12-Layer Encoder-Decoder
Token Modification	0% Change	Autoregressive Syntactical Synthesis
Grammar Correction	Disabled	Enabled via Cross-Attention
Output Sequence (Y)	$Y = G$	$Y = \text{Argmax}(P(Y G))$

4 Experiments

4.1 Data

This study used the GKSL3k dataset [9], which contains paired Korean Sign Language gloss sentences and Korean natural language sentences. The dataset consists of 3,052 sentence pairs. Each sample includes a Gloss level Korean Sign Language (GKSL) sentence and its corresponding Korean sentence.

For this project, the task was defined as a sequence-to-sequence generation problem where the model translates gloss sequences into grammatically correct Korean sentences. The gloss sequence was used as the input, while the Korean sentence was used as the target output.

Before training, the dataset was preprocessed by removing empty samples and normalizing whitespace. Since the task requires fluent Korean sentence generation, Korean punctuation marks such as periods and question marks were preserved. The gloss tokens were also kept space-separated to maintain token-level semantic structure.

To improve sentence-type generation, an additional sentence tag was added to the input gloss sequence. Sentences ending with a question mark were labeled with the [QUESTION] token, while all other sentences were labeled with the [STATEMENT] token. The tag was added as a prefix to the gloss input sequence.

Input : [QUESTION] 너 어디 가다
Output: 너는 어디 가니?

The dataset was divided into training, validation, and test sets using an 80:10:10 ratio. To prevent data leakage, grouped splitting was applied based on gloss sentences so that identical gloss inputs did not appear across multiple splits.

The final dataset split consisted of approximately:

Split	Samples	Percentage
Train	2396	80%
Validation	318	10%
Test	338	10%

Table 2: Dataset split of GKSL3k

4.2 Evaluation Method

The performance of the proposed gloss-to-Korean translation model was evaluated using automatic text generation metrics and qualitative analysis. Since the task involves generating natural Korean sentences from Gloss level Korean Sign Language (GKSL) sequences, the evaluation focused on semantic similarity, lexical overlap, and sentence fluency.

Three evaluation metrics were used: BLEU, ROUGE-L, and BERTScore.

BLEU (Bilingual Evaluation Understudy) measures the n-gram overlap between the generated sentence and the reference sentence. BLEU is widely used in machine translation tasks because it evaluates how closely the generated output matches the target sentence at the lexical level. Higher BLEU scores indicate better translation quality.

ROUGE-L evaluates the longest common subsequence between the generated sentence and the reference sentence. Unlike BLEU, ROUGE-L focuses more on sequence-level similarity and sentence structure preservation. This metric is particularly useful for evaluating sentence generation tasks where word ordering and fluency are important.

BERTScore was additionally used to measure semantic similarity between generated and reference sentences. BERTScore computes contextual embedding similarity using pretrained language representations, allowing semantically similar sentences to receive higher scores even when exact word overlap is low. This is important for gloss-to-text generation because multiple valid Korean sentences can correspond to the same gloss sequence.

In addition to automatic metrics, qualitative evaluation was conducted through manual inspection of generated outputs. The analysis focused on:

- grammatical correctness,
- sentence fluency,
- appropriate particle generation,
- sentence-type generation (question vs. statement),
- semantic consistency with the input gloss sequence.

The validation dataset was monitored during training to reduce overfitting. Early stopping was applied when the validation loss did not improve for two consecutive evaluation steps.

4.3 Experimental Details

The experiments were conducted using KoBART, a pretrained Korean sequence-to-sequence language model based on the BART architecture. The implementation used the HuggingFace Transformers framework together with the KoBART tokenizer and pretrained weights.

The task was formulated as a sequence-to-sequence translation problem where the input consists of GKSL gloss sentences and the output consists of Korean natural language sentences. Intent-tag preprocessing follows the procedure described in Section 4.1.

The dataset was preprocessed through whitespace normalization and removal of empty samples. The gloss tokens were preserved as space-separated units, while Korean punctuation was retained to support fluent sentence generation.

The dataset was divided into training, validation, and test sets using an 80:10:10 split ratio. Grouped splitting based on gloss sentences was applied to prevent identical gloss inputs from appearing across multiple splits, reducing data leakage during evaluation.

The model used in the experiments was `gogamza/kobart-base-v2`. Training was performed using the AdamW optimizer with mixed precision training enabled when CUDA was available.

The training configuration is summarized below:

- Model: `gogamza/kobart-base-v2`
- Maximum input length: 64
- Maximum target length: 64
- Batch size: 8
- Learning rate: 3×10^{-5}
- Number of epochs: 5
- Weight decay: 0.01
- Early stopping patience: 2
- Random seed: 42

The HuggingFace Seq2SeqTrainer framework was used for training and validation. Evaluation and checkpoint saving were performed at the end of every epoch. The best model was selected based on validation loss using the `load_best_model_at_end` option.

During inference, sequence generation was performed using autoregressive decoding with a maximum generation length of 40 tokens. The trained tokenizer and best-performing model checkpoint were saved after training for final evaluation on the test set.

4.4 Results

The experimental results demonstrate that the pretrained KoBART model was capable of generating understandable Korean sentences from GKSL gloss inputs despite the relatively small dataset size.

Table 3 presents the quantitative evaluation results of the proposed gloss-to-Korean translation model.

Table 3: Performance of the Proposed KoBART Model on the GKSL Gloss-to-Korean Translation Task

Model	BLEU	ROUGE-L (F1)	BERTScore (F1)
KoBART (ours)	66.46	0.7253	0.9231

The model achieved a BLEU score of 66.46, indicating strong lexical similarity between generated sentences and reference sentences. The ROUGE-L F1 score of 0.7253 demonstrates that the generated outputs preserved sentence-level structure and word ordering effectively.

Furthermore, the model achieved a BERTScore F1 of 0.9231, suggesting high semantic similarity between generated and reference sentences. This result indicates that the model was capable of producing semantically accurate Korean sentences even when exact word overlap differed.

The model successfully learned basic grammatical structures and sentence-type generation. Example outputs indicate that the system recovered omitted grammatical particles and generated fluent Korean sentences from simplified gloss representations.

Input Gloss: [STATEMENT] 나 학교 가다
Generated Sentence: 나는 학교에 갔다.

However, several limitations were observed. The model occasionally generated incorrect particles, repetitive words, or simplified sentence structures when handling longer or more complex gloss inputs. Performance was also limited by the relatively small dataset size and the variability of gloss expressions.

Overall, the experiments demonstrate that fine-tuning a pretrained Korean sequence-to-sequence model is an effective approach for gloss-to-text translation in Korean Sign Language, even with limited training data.

4.5 Discussion of Design Choice

Intent tags ([STATEMENT], [QUESTION]) were introduced to reduce ambiguity between declarative and interrogative sentence generation. Although ablation was outside the project scope, qualitative observations suggest improved sentence-type consistency.

5 Analysis

A qualitative evaluation was conducted by manually examining the generated outputs and comparing them with the reference sentences. The analysis aims to understand the strengths and weaknesses of the proposed system beyond numerical evaluation metrics.

Overall, the model performs well on short and direct gloss sequences, especially for common transportation and emergency-related expressions. In several cases, the generated output exactly matches the reference sentence.

Expected Output	Generated Output
파리바게트 지나서 내려주세요.	파리바게트 지나서 내려주세요.
편의점은 어디에 있나요?	편의점은 어디에 있나요?

Table 4: Examples of correct predictions

The model is also capable of preserving semantic meaning even when the wording differs.

Expected Output	Generated Output
사람이 갑자기 쓰러졌어요.	어떤 사람이 갑자기 쓰러지신 것 같아요.

Table 5: Example of semantic preservation

However, several failure cases were observed. Some outputs become semantically unrelated to the input gloss sequence.

Expected Output	Generated Output
몸이 결박 당해있어요.	이 주소로 가주세요.

Table 6: Example of semantic mismatch

The model also occasionally produces awkward or unnatural sentences.

Expected Output	Generated Output
빨리 가주세요.	운전하다 빠르다.

Table 7: Example of awkward sentence generation

From the qualitative analysis, it can be observed that the model performs better on simple and frequently occurring expressions, while performance decreases for ambiguous or less common gloss sequences. Overall, the system demonstrates the ability to generate understandable Korean sentences, although improvements are still needed in contextual understanding and sentence fluency.

6 Conclusion

In this project, we proposed a gloss-to-text generation framework that converts Korean Sign Language (KSL) gloss sequences into natural Korean sentences using KoBART. Unlike previous studies that primarily focused on sign recognition from video inputs or gloss extraction, our work addressed the subsequent language generation stage by formulating the problem as a sequence-to-sequence translation task.

To improve sentence generation quality, we introduced intent-based input conditioning using [STATEMENT] and [QUESTION] tags to explicitly represent sentence type information during training and inference. The model was trained on the GKSL3k dataset and optimized using a pretrained Korean encoder-decoder transformer architecture.

Experimental evaluation demonstrated the effectiveness of the proposed approach. The model achieved a BLEU score of 66.46, ROUGE-L F1 score of 0.7253, and BERTScore F1 score of 0.9231, indicating strong lexical, structural, and semantic alignment between generated and reference sentences. Qualitative analysis further showed that the system successfully recovered omitted grammatical structures and produced understandable Korean sentences from simplified gloss inputs.

Overall, the results demonstrate that pretrained Korean sequence-to-sequence language models can effectively bridge the linguistic gap between gloss representations and natural language generation under limited data conditions.

7 Future Work

Although the proposed model achieved encouraging results, limitations remain in handling ambiguous gloss inputs, long sequences, and grammatical consistency. Errors such as incorrect particles, semantic mismatch, and repetitive generation were still observed.

Future work may improve performance by expanding dataset diversity, introducing richer linguistic annotations, and incorporating context-aware generation and advanced decoding strategies. Extending the framework into an end-to-end Korean Sign Language translation pipeline is also a promising direction.

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A Appendix: Team contributions

A.1 Shahmie Nur Ardini Shaharudin (Leader)

- **Project Coordination:** Managed overall project workflow, task distribution, timeline planning, and final integration.
- **Code:** Assisted KoBART training pipeline setup, GitHub repository structure organization, and AWS/GPU environment coordination.
- **Report:** Abstract, Introduction, Related Work, Conclusion, and Appendix sections. Also performed final report integration and slide preparation.

A.2 Muhammad Afiq Zulhusni Zailani (Member)

- **Code:** Implemented the main KoBART fine-tuning pipeline, model loading, hyperparameter tuning, training experiments, and decoding configuration.
- **Report:** Approach section including system architecture, KoBART fine-tuning process, and training methodology.

A.3 Muhammad Aliff Mohd Nawawi (Member)

- **Code:** Collected gloss-sentence datasets, performed dataset preprocessing, formatting, tokenization preparation, train/validation/test splitting, and dataset variant construction.
- **Report:** Experiments section including dataset and preprocessing descriptions.

A.4 Nur Irdina Zahari (Member)

- **Code:** Implemented evaluation metrics including BLEU, ROUGE, and BERTScore, performed dataset verification, generated evaluation outputs, and conducted error analysis.
- **Report:** Analysis section including evaluation interpretation, qualitative analysis, error analysis, and experimental result discussion.